



**Maynooth
University**

National University
of Ireland Maynooth

**SEMESTER 1
2023-2024**

**CS370FZ
Computation and Complexity**

Dr. Niu Yuzhen, Dr. Joe Timoney, Dr. Michael English

Time allowed: 2 hours

Answer all *three* questions

All questions carry equal marks

Instructions

	Yes	No
Log Books Allowed	☐	☒
Formula Tables Allowed	☐	☒
Other Allowed (<i>enter details</i>)	☐	☒

General (*enter detail*)

QUESTION 1

- (a) Give self-contained definitions of the following languages/Turing machines: (12 marks)
- (i) TM M decides a Language L
 - (ii) A_{TM} (the acceptance problem)
 - (iii) Nondeterministic TM M semidecides a Language L
- (b) Based on the following copy Turing machine, give the sequence of tape changes on input $\sqcup aa \sqcup$. (5 marks)
- (c) Prove the following language is not recursively enumerable. (8 marks)
- $$\overline{H_1} = \{w : \text{either } w \text{ is not the encoding of a Turing machine, or it is the encoding "M" of a Turing machine } M \text{ that does not halt on "M"}\}.$$

QUESTION 2

- (a) State the Church-Turing Thesis. (4 marks)
- (b) State the Rice's theorem. (5 marks)
- (c) Prove that Rice's Theorem. (You are required to use the recursion theorem) (8 marks)
- (d) Prove that A_{TM} is undecidable. (You are required to use the diagonalization technique) (8 marks)

QUESTION 3

- (a) Choose **five** terms of the following sets and give their definitions. Your definition should be self-contained (e.g. for CLIQUE, explain what a k-clique is). (10 marks)
- (i) PATH
 - (ii) RELPRIME
 - (iii) SAT
 - (iv) HAMPATH
 - (v) COMPOSITES
 - (vi) CLIQUE
- (b) Prove that CLIQUE is in NP. (5 marks)
- (c) Recall the proof 3SAT is polynomial time mapping reducible to VERTEX-COVER ($3SAT \leq_P \text{VERTEX-COVER}$). From the following ϕ construct the associated graph G . Explain how to obtain an 8-vertex-cover for G from a satisfying assignment for ϕ . (10 marks)

$$\phi = (\bar{X}_1 \vee X_2 \vee X_2) \wedge (\bar{X}_1 \vee \bar{X}_2 \vee \bar{X}_2) \wedge (X_1 \vee \bar{X}_2 \vee \bar{X}_2)$$